

Advanced (Habanero)

Q1. Solve for x : $2^x = 8$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. Solve for y : $3^y = 27$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Q3. Solve for z : $4^z = 64$

- (A) $z = 1$
- (B) $z = 2$
- (C) $z = 3$
- (D) $z = 4$
- (E) $z = 5$

Q4. Solve for x : $x^2 = 25$

- (A) $x = 3$
- (B) $x = 4$
- (C) $x = 5$
- (D) $x = 6$
- (E) $x = pm5$

Q5. Solve for y : $y^3 = 125$

- (A) $y = 4$
- (B) $y = 5$
- (C) $y = 6$
- (D) $y = 7$
- (E) $y = 8$

Q6. Solve for z : $z^4 = 256$

- (A) $z = 2$
- (B) $z = 3$
- (C) $z = 4$
- (D) $z = 5$
- (E) $z = 6$

Q7. Solve for x : $2^{x+1} = 16$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 3$

Q8. Solve for y : $3^{y-1} = 9$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 2$

Open Ended Questions (FRQ)

Q9. Solve for x : $x^2 = 4^x$

Q10. Solve for y : $y^3 = 2^{y+1}$

Chef's Tips

- Read each question carefully
- Use the back of the page for calculations
- Chef's Tips